Applications in AI and Data Science

Coursework Report: Property Price Analytics and Facial Expression Detection

COMP1891 (2025/2026)

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Abstract

This report applies a complete data science and machine learning workflow to two business scenarios: a 350-record estate agency dataset used to model residential property prices, and a sample of 50 facial images supporting a workplace wellbeing initiative. All results are produced by the accompanying notebook `COMP1891_notebook.ipynb`, whose cell headings match the section numbers used here.

Contents

1 Section A: Data Science (Scenario 1)

1.1 A1 – Exploratory Data Analysis

The supplied file `Estate_Agent.csv` contains 350 records and nine columns. It is semicolon-delimited rather than comma-delimited, so the loader detects the separator from the header line instead of assuming a default; read with the wrong separator the file collapses into a single unusable column.

The nine columns divide into three roles. `House_Price` is the regression target. `Rating` is an ordinal quality grade used as the classification target in Section B. The remaining seven – `House_Area`, `Build` (year of construction), `No_of_Bathrooms`, `No_of_Bedrooms`, `Garage_Area`, `Living_Space` and `Parking` – are predictors describing the physical property.

All seven predictors are retained, on two tests: does the feature describe the property, and does it carry signal? Table 1 shows every predictor correlating with price at $|r| \geq 0.15$, and none is a proxy for a protected characteristic – there is no postcode, neighbourhood or demographic column, which materially reduces the discrimination risk discussed in Section D. `Rating` is deliberately excluded: it correlates with price at $r = -0.045$, and is a subjective assessment rather than a physical measurement, so including it would make the model depend on a human judgement the agency wants to benchmark.

Table 1: Pearson correlation with `House_Price` (after cleaning).

Feature	r **with House_Price**
Parking	0.691
Garage_Area	0.680
Living_Space	0.608
No_of_Bathrooms	0.531
Build	0.531
House_Area	0.286
No_of_Bedrooms	0.156
Rating	−0.045

The strongest predictor is `Parking` ($r = 0.691$), narrowly ahead of `Garage_Area` ($r = 0.680$). Both measure the same underlying attribute – garage capacity as a count and as an area – and are strongly correlated with each other, which matters for interpreting the regression coefficients in Section B1.

The most commercially significant finding is the near-zero correlation between `Rating` and price: the agency's own quality grade does not predict what a property sells for. Either the grade captures something buyers do not pay for, or it is applied inconsistently, and both readings recommend reviewing the grading process before relying on it for pricing strategy. Figure 1 presents the full correlation matrix.

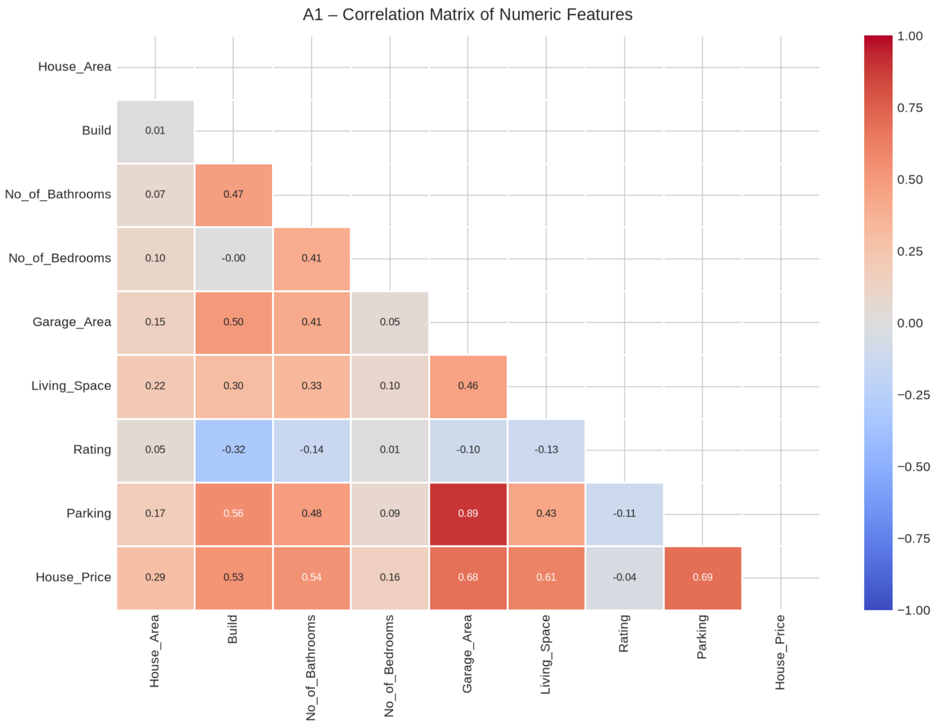

Figure 1: A1 – Correlation matrix of all numeric features.

1.2 A2 – Data Cleaning

The raw data is not analytics-ready. Four distinct defects were found and corrected.

Formatting errors in a numeric column. `Parking` is stored as text and mixes digits with written words: nine records hold `One` (4), `Two` (3) or `Three` (2). A naive conversion turns all nine into missing values, which median imputation then silently overwrites with plausible-looking numbers. Since `Parking` is the strongest price predictor, this is the most damaging defect in the file. The cleaning routine maps written numbers to integers before conversion and reports what it changed.

Duplicate records. Four exact duplicates were removed (350 → 346). They bias every statistic and can leak the same property into both training and test sets.

Missing values. Four columns had gaps: `Living_Space` (5), `Garage_Area` (3), `No_of_Bathrooms` (2) and `No_of_Bedrooms` (2), never more than 1.4% of any column. Median imputation was used rather than mean because the area columns are right-skewed, and the median is not dragged by the upper tail.

Outliers. Treatment was matched to each column rather than applied as one blanket rule (Table 2). Price uses the conventional 1.5×IQR fence, excluding 14 records. The area columns are naturally right-skewed – large plots are genuine, not erroneous – so the same rule would discard 17 valid properties from `House_Area` alone. A 3×IQR fence was used there, removing only the 5 implausible extremes, including one plot of 159,000 square feet against a median of 9,378. The data falls from 346 to 327 rows.

Table 2: Outlier audit. The applied rule is shown in bold.

Column	Fence	Lower	Upper	Outliers	% of rows
House_Price	**1.5 IQR**	688	343,188	**14**	4.0
House_Price	3.0 IQR	−127,750	471,625	2	0.6
House_Area	1.5 IQR	1,603	17,569	17	4.9
House_Area	**3.0 IQR**	−4,384	23,556	**6**	1.7
Living_Space	1.5 IQR	173	2,075	5	1.4
Garage_Area	1.5 IQR	−63	959	3	0.9

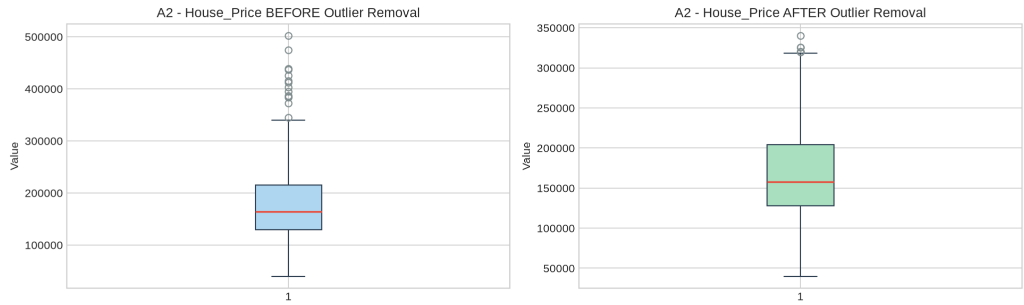

Figure 2: A2 – Price distribution before and after outlier removal.

1.3 A3 – Data Visualisation

Each visualisation was chosen to answer a specific question rather than to display the data generically.

Price distribution (Figure 3). Linear regression assumes approximately normal residuals, and a histogram paired with a Q–Q plot tests that assumption directly. Post-cleaning skewness is 0.707 – below the |skew| > 1 threshold at which a log transformation would normally be applied, so price was modelled untransformed.

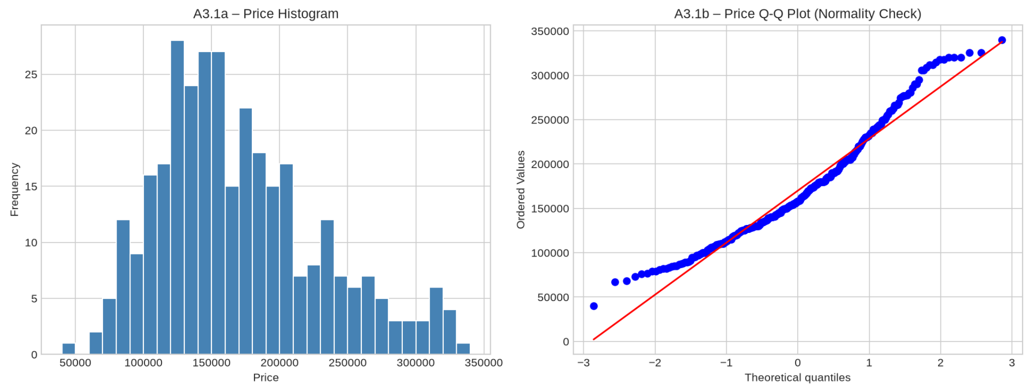

Figure 3: A3.1 – Price histogram and Q–Q plot.

Feature distributions (Figure 4). Count and continuous variables are plotted differently. Bedrooms, bathrooms, rating and parking take only a handful of whole-number values, so drawing them as histograms creates meaningless fractional bins and hides how uneven the categories are; they are drawn as category bars annotated with the share held by the largest level, while continuous features keep histograms annotated with skewness.

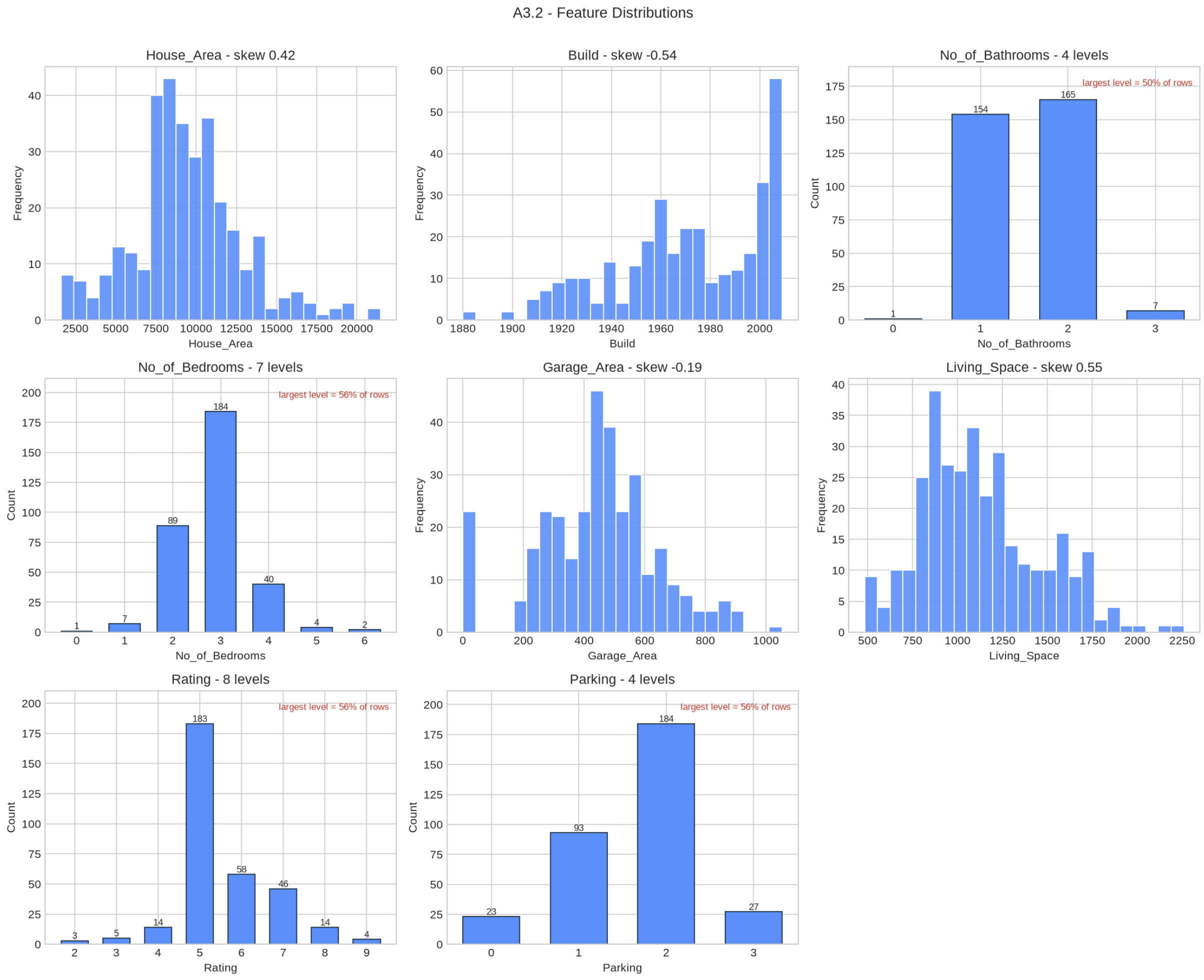

Figure 4: A3.2 – Feature distributions, split by variable type.

Price against the leading predictors (Figure 5). Scatter plots with a fitted trend line test whether the relationship that the correlation coefficient summarises is genuinely linear, since a single correlation value cannot distinguish a linear relationship from a curved one.

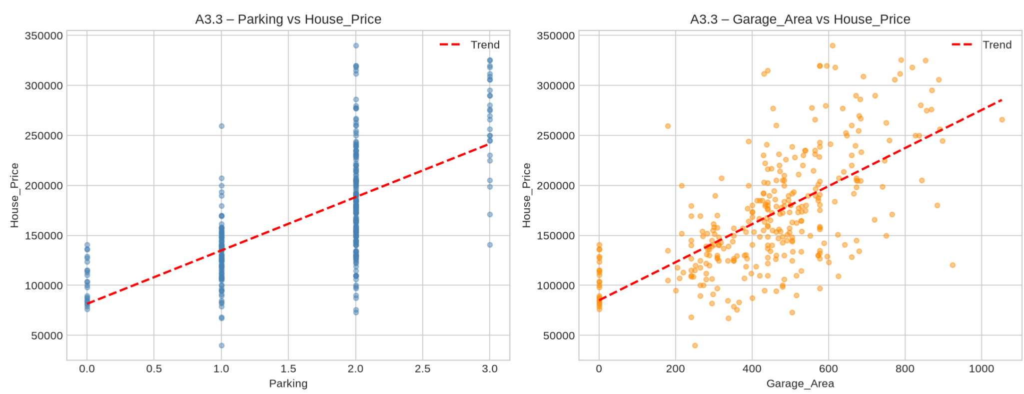

Figure 5: A3.3 – Price against the two most correlated features.

Price by category (Figure 6) shows how price disperses across grades and bedroom counts, with group sizes printed on the axis so that a wide box resting on three properties is not mistaken for a reliable pattern. It confirms visually what Table 1 states numerically: median price does not climb consistently with rating.

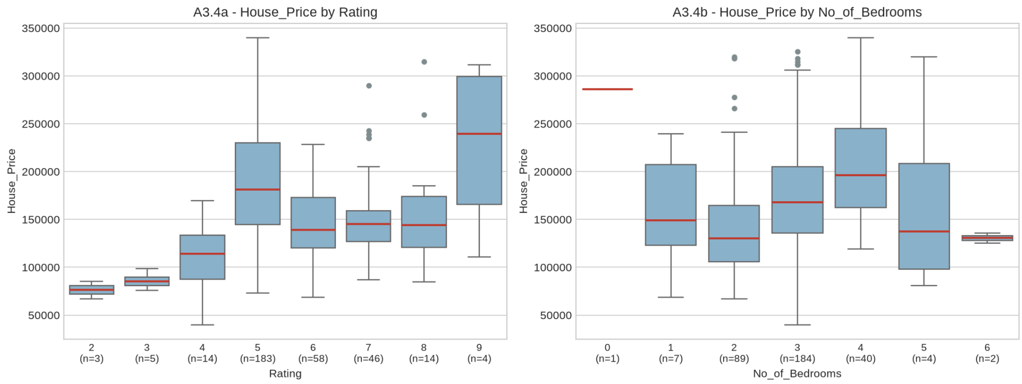

Figure 6: A3.4 – Price by rating and by bedroom count (n shown per group).

Class balance (Figure 7). Because `Rating` is the target in Sections B2 and B3, its balance was examined before any model was fitted. Grade 5 holds 56.0% of properties, the imbalance ratio is 61:1, and grades 2, 3 and 9 have fewer than ten examples each. This chart determines the choice of metric throughout Section B: a model predicting grade 5 unconditionally already scores 56% accuracy while learning nothing.

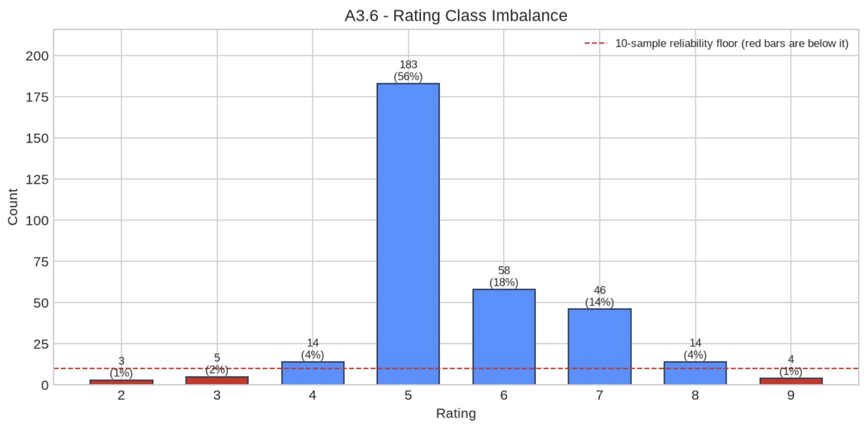

Figure 7: A3.6 – Rating class imbalance. Red bars fall below a usable sample size.

2 Section B: AI and Machine Learning (Scenario 1)

All models use the same seven predictors, standardised with `StandardScaler`. Scaling is essential for kNN and SVM, which measure distances, and harmless for linear regression.

2.1 B1 – Multiple Linear Regression

Three train/test splits were evaluated (Table 3). Three metrics are reported together because each answers a different question: MAE gives the average error in pounds and is what a client would understand; RMSE penalises large errors more heavily and so exposes occasional bad predictions; R^2 states the proportion of price variance explained.

Table 3: B1 – Linear regression across three train/test splits.

Split	Train	Test	MAE	RMSE	R^2
70/30	228	99	23,702.69	31,972.23	0.76
80/20	261	66	24,111.50	31,917.46	0.76
90/10	294	33	23,772.38	30,314.87	0.81

The model explains 76–81% of price variance with a typical error near £24,000 on a median price of roughly £164,000 – around 15%, useful for initial guidance but not precise enough to set an asking price unaided. Performance is stable across splits, indicating the model is not sensitive to how the data is divided.

The 90/10 split reports the highest R^2 (0.81), but this is not the best model: its test set holds only 33 properties, making it the least reliable estimate of the three, and the apparent gain is a small-sample artefact. The 70/30 and 80/20 results, agreeing at $R^2 = 0.76$, are the more trustworthy estimate of real-world performance.

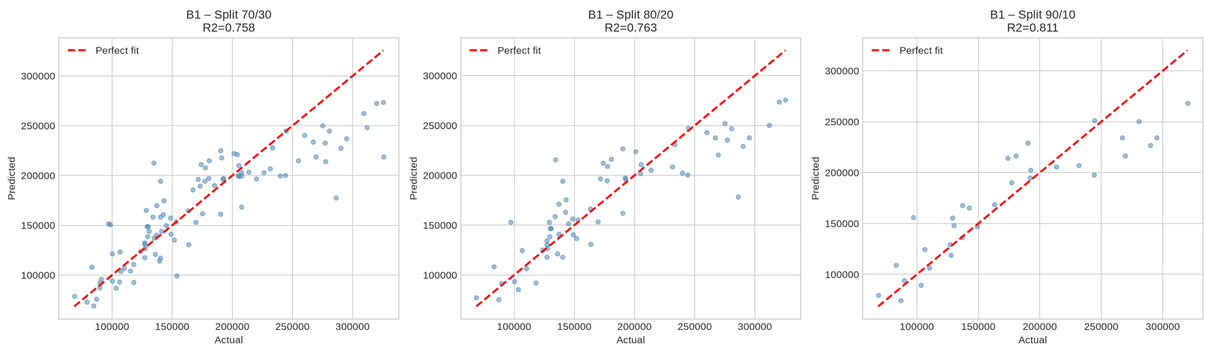

Figure 8: B1 – Predicted against actual price for each split.

Figure 9 shows the fitted coefficients. These must be read with care: as noted in A1, `Parking` and `Garage_Area` both measure garage provision and are strongly correlated, so the regression distributes credit between them somewhat arbitrarily. Their individual coefficients are therefore not a safe guide to the value of an extra parking space, even though their combined contribution to prediction is sound.

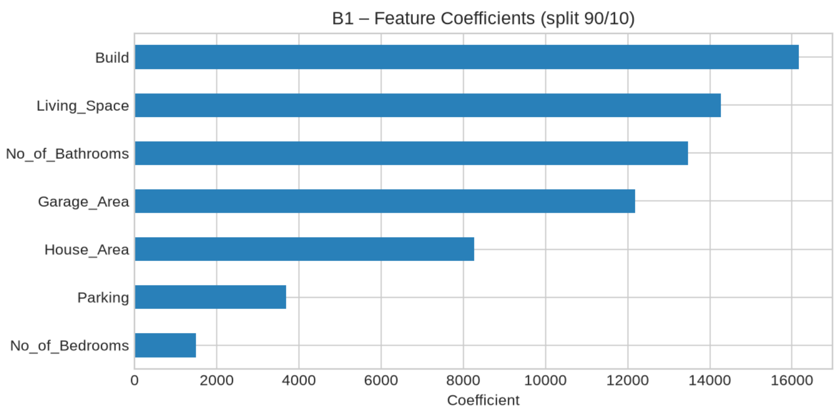

Figure 9: B1 – Standardised feature coefficients.

Applying the model to a median property (9,180 sq ft plot, built 1972, 2 bathrooms, 3 bedrooms, 454 sq ft garage, 1,085 sq ft living space, 2 parking spaces) predicts **£181,951**. Given the MAE, this should be quoted to a client as approximately £182,000 ± £24,000 rather than as a point figure.

2.2 B2 – k-Nearest Neighbours

k was tuned across $k = 1 \dots 25$ using stratified cross-validation on the training set only. Selecting k by test-set accuracy – a common shortcut – leaks the test data into model selection and produces an optimistic score. The difference is visible here: cross-validation selects $k = 11$, whereas $k = 7$ maximises test accuracy. Only the former is a legitimate choice.

The extreme imbalance forces a compromise. The smallest training class holds just two properties, capping the cross-validation at two folds.

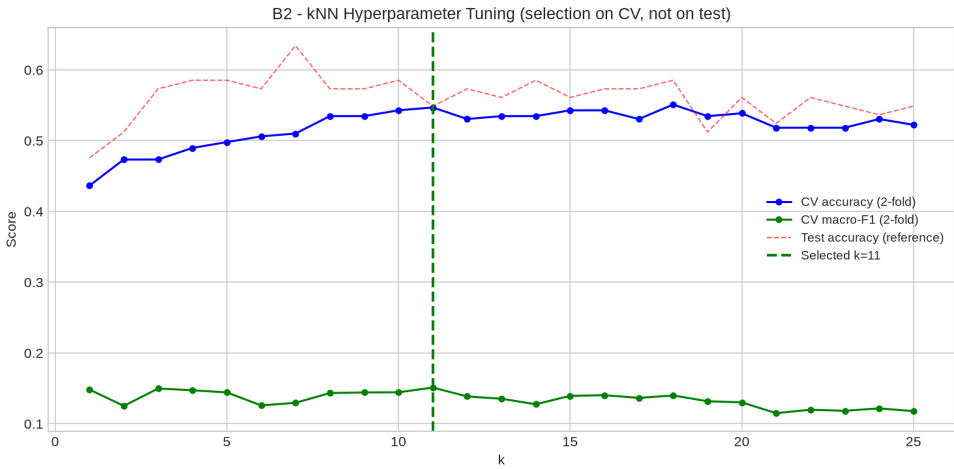

Figure 10: B2 – kNN tuning. Selection uses the cross-validated curve, not the test curve.

At $k = 11$ the model achieves 54.9% accuracy – *below* the 56.1% majority-class baseline. Balanced accuracy is 0.155 and macro-F1 0.150. The per-class report shows why: grades 2, 3, 4, 8 and 9 all score zero on precision and recall. The model has effectively learned to predict grade 5 and little else. Reporting only weighted F1 (0.491) would disguise this completely, which is precisely why macro-averaged metrics are reported alongside.

2.3 B3 – Support Vector Machine

Three kernels were tuned over C (and γ) by cross-validation, with `class_weight='balanced'` so that rare grades are not discarded in favour of the dominant one (Table 4).

Table 4: B3 – Tuned, class-balanced SVM by kernel.

Kernel	Best params	CV macro-F1	Accuracy	Balanced acc.	Macro-F1
Linear	$C = 100$	0.154	0.463	0.283	0.214
RBF	$C = 100$, $\gamma = 0.1$	**0.177**	0.512	**0.332**	**0.323**
Poly (3)	$C = 1$, γ=scale	0.172	0.524	0.317	0.253

The kNN/SVM comparison depends entirely on which metric is chosen. kNN wins on raw accuracy (0.549 vs 0.512); the RBF SVM wins decisively on balanced accuracy (0.332 vs 0.155) and macro-F1 (0.323 vs 0.150). Class weighting causes the SVM to trade errors on the dominant grade for coverage of the rare ones, which lowers accuracy while more than doubling macro-F1. Since the agency's interest lies in identifying unusually high- and low-grade properties rather than in restating the most common grade, the RBF SVM is the better model despite its lower headline accuracy.

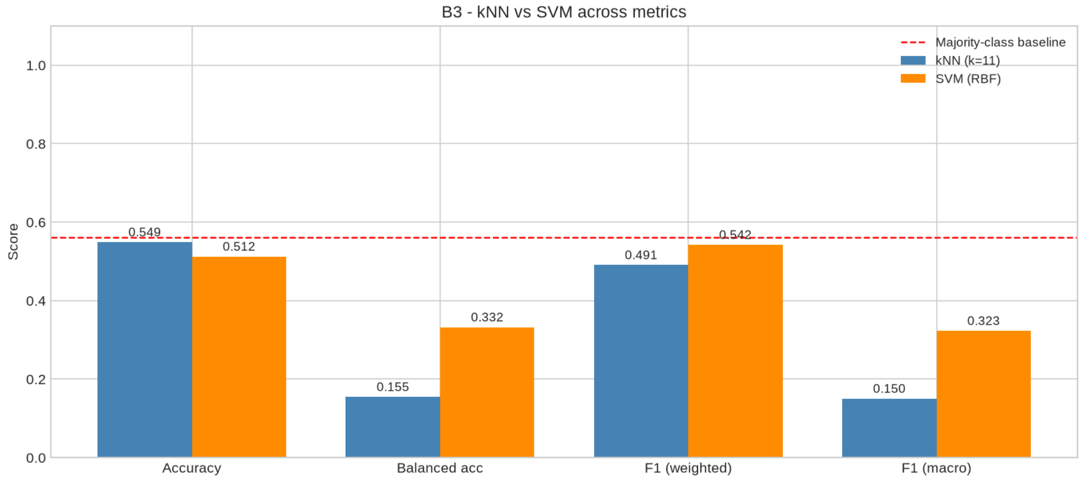

Figure 11: B3 – kNN against SVM across four metrics, with the majority-class baseline.

Addressing the imbalance. Neither model beats the baseline convincingly, because grades with three to five examples cannot be learned. Grouping the eight grades into three commercially meaningful bands – Low (≤ 4, $n = 22$), Mid (5–6, $n = 241$) and High (≥ 7, $n = 64$) – keeps the business question intact while giving every class usable support, and restores full 5-fold cross-validation. Macro-F1 rises from 0.323 to 0.503 (Table 5).

Table 5: B3.3 – Banded rating target (baseline accuracy 0.732).

Model	Accuracy	Balanced acc.	Macro-F1
kNN ($k = 5$)	0.683	0.422	0.436
SVM (RBF, $C = 10$, $\gamma = 0.1$)	0.634	**0.561**	**0.503**

The improvement does not come from a smarter model; it comes from asking the question at a resolution the data can support. An estate agent advised on whether a property is a low, mid or high grade asset is well served. A claim to separate grade 8 from grade 9 on three examples is not defensible.

2.4 B4 – Clustering and Dimensionality Reduction

k-means was run for $k = 2 \ldots 10$ and evaluated by the elbow method and silhouette score. The silhouette peaks at $k = 2$ (0.287), giving two balanced segments of 161 and 166 properties – broadly a smaller/older and a larger/newer segment.

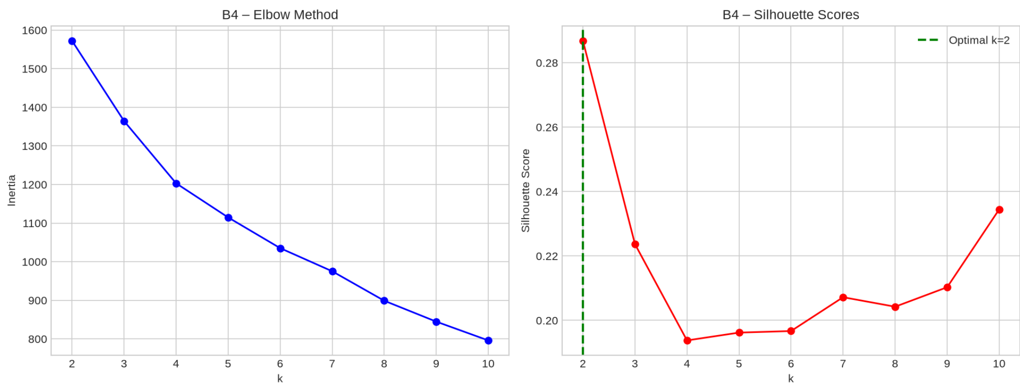

Figure 12: B4 – Elbow and silhouette curves.

Both PCA and t-SNE were applied to project the seven features into two dimensions. Rather than asserting which performs better, both were measured (Table 6).

Table 6: B4 – PCA against t-SNE.

Space	Silhouette	Variance retained	Interpretable axes	Deterministic
Original (7-D)	0.287	100%	yes	yes
PCA (2-D)	0.420	61.6%	yes	yes
t-SNE (2-D)	**0.572**	n/a	no	no

t-SNE produces visually tighter groups (silhouette 0.572 against 0.420), but its axes carry no units, between-cluster distances are not meaningful, and the layout changes with the seed and perplexity. PCA retains 61.6% of variance in two components with axes interpretable as weighted combinations of the original features. PCA is therefore the more appropriate estimator here, because the agency must explain to a client *why* a property falls into a segment; t-SNE remains better for the narrower task of confirming visually that two separable segments exist.

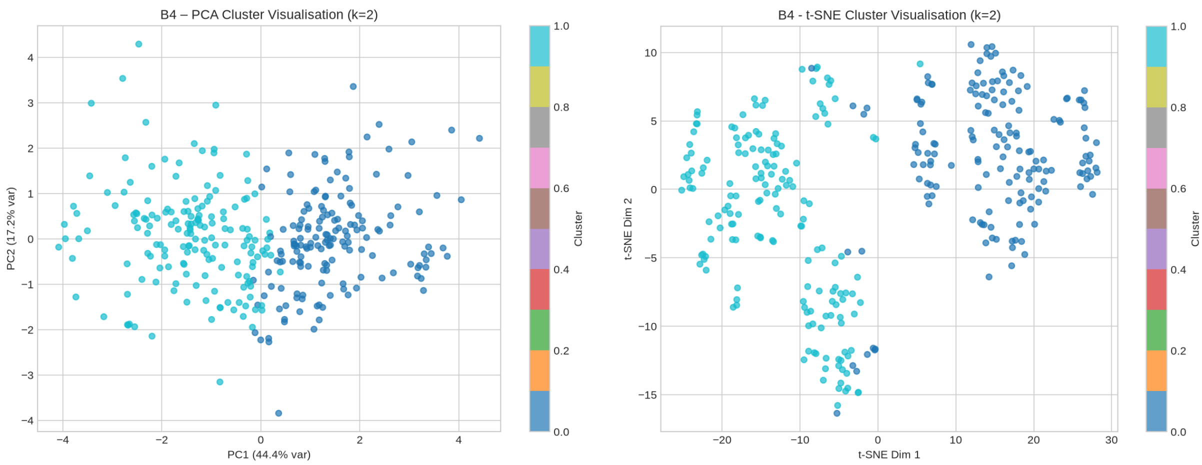

Figure 13: B4 – Clusters under PCA (left) and t-SNE (right).

3 Section C: AI – Facial Expression Detection (Scenario 2)

3.1 C1 – Detection and Measured Accuracy

Fifty images in `FacesSample` were classified into the six expressions named in the brief: Surprised, Angry, Happy, Sad, Frightful and Neutral. An `Expressions` folder holding one self-selected reference image per expression was created as required. Crucially, the sample ships with ground-truth labels (`groundtruth.json`), so detection success is *measured* rather than estimated by eye.

Two methods were implemented and compared. Both use `trpakov/vit-face-expression`, a Vision Transformer fine-tuned on facial expressions.

- **Method 1 – reference matching.** Every image is represented by its 768-dimension CLS embedding, and each sample image takes the label of the most cosine-similar reference image. This uses the `Expressions` folder as the task specifies, and is one-shot learning: a single exemplar per class.

- **Method 2 – direct classification.** The same network's 7-way softmax head predicts the expression directly, having been trained on thousands of labelled faces. Its `disgust` output has no counterpart in the brief and is folded into Angry.

Table 7: C1 – Detection accuracy against ground truth ($n = 50$).

Method	Accuracy	Balanced acc.	Macro-F1
Random guess (6 classes)	0.167	–	–
Majority class	0.260	–	–
Method 1 – reference match	0.660	0.592	0.570
Method 2 – ViT classifier	**0.720**	**0.708**	**0.684**

How successful is the detection? Both methods substantially beat the 26.0% majority-class baseline, so both genuinely detect expression. Method 2 reaches 72.0% accuracy; Method 1 reaches 66.0%, a strong result for learning each expression from exactly one example, confirming that the embedding space groups faces by expression rather than by identity or lighting. The 6-point gap is the price of one-shot learning: with a single reference per class, an unrepresentative exemplar harms an entire category. Method 1 scores 0.00 F1 on Frightful, failing all four frightful faces because its single reference does not resemble them, while Method 2 recovers them at 0.75 recall.

Are any images wrongly detected as Neutral? The methods fail in opposite directions. Method 1 never wrongly assigns Neutral, but misses seven of the nine true Neutral faces – recall of just 0.22. Method 2 wrongly labels two images Neutral, `fig109.jpg` (truly Angry) and `fig145.jpg` (truly Sad), and misses six true Neutrals, for 0.33 recall.

Neutral is the hardest class for both, which is intuitive: it is defined by the *absence* of expression, so any weak or partial expression sits close to it, and a low-intensity angry or sad face is genuinely near-neutral in appearance. This matters for the wellbeing application, where misreading distress as neutral is exactly the error of greatest consequence.

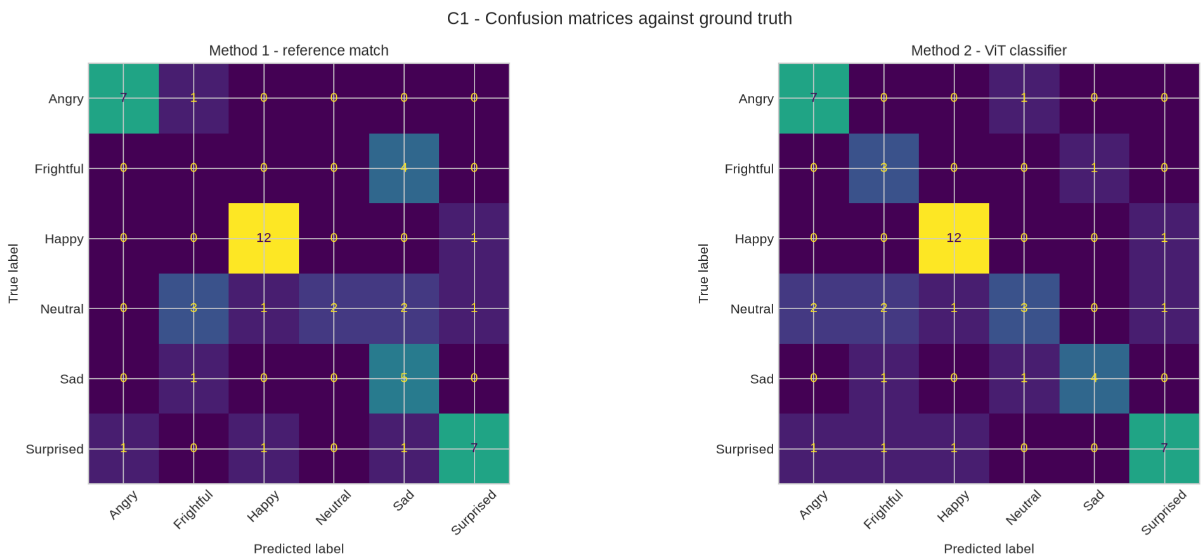

Figure 14: C1 – Confusion matrices for both methods against ground truth.

Prediction confidence is a usable safeguard. Method 2 is 72.9% accurate when its confidence is at least 0.40, but only 50.0% below it. Low-confidence predictions can therefore be routed for human review rather than acted on automatically.

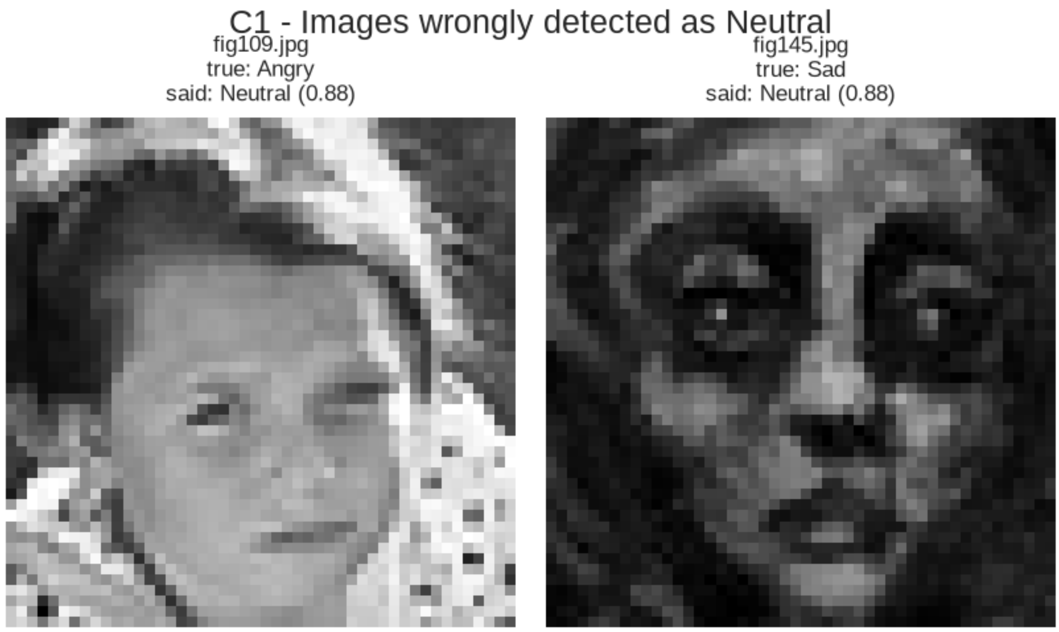

Figure 15: C1 – Images wrongly detected as Neutral.

3.2 C2 – Application: Aggregated Wellbeing Monitoring

C1 turns a face into a label. C2 models the operational system that would sit on top of it for the corporate health and wellness initiative. Expressions are mapped to a stress weight (Happy 0.0 through Angry 0.9), averaged per employee per week over four weeks, and screened by two rules: a *level* rule (weekly mean above 0.55) and a *trend* rule (a rise above 0.15 from the first week to the last).

Two rules are needed because they catch different problems. The level rule finds employees who are consistently under strain; the trend rule catches deterioration early, flagging employees whose absolute score is still moderate but climbing. In the simulation the trend rule flags two employees rising by 0.32 and 0.36 whom the level rule alone would have missed until later.

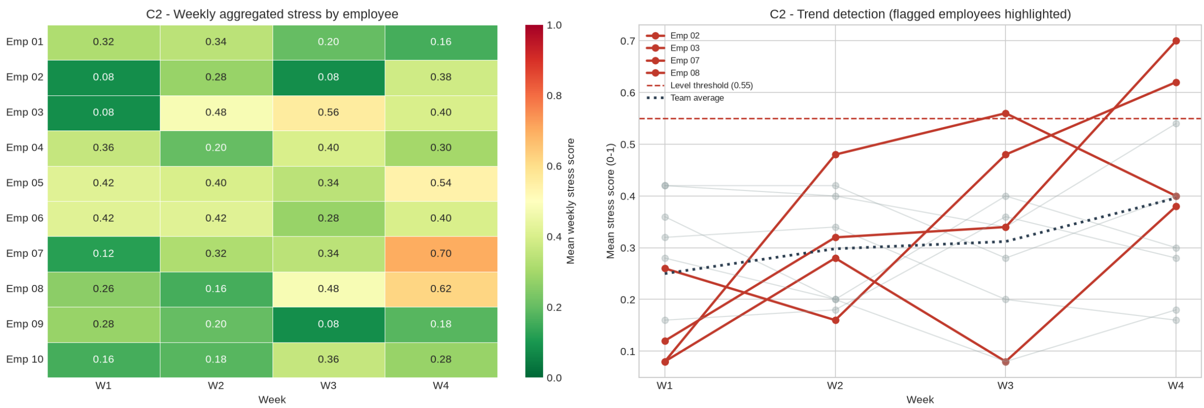

Figure 16: C2 – Weekly aggregated stress and trend detection.

The design is constrained by the ethics discussed in Section D and by the measured accuracy in C1. Rules operate on weekly aggregates, never on individual daily readings; managers see the team average and the number of employees flagged, not the per-person matrix; participation is opt-in and withdrawable. Because C1 measured only 72% accuracy with Neutral the weakest class, a flag prompts an *offer* of a voluntary check-in and never an automated intervention or any input to performance management.

4 Section D: AI and LSEPI

4.1 D1 – Legal, Social, Ethical and Professional Issues

Scenario 1 – price prediction. *Legally*, property records combined with addresses are personal data under the UK GDPR (ICO, 2018); this dataset avoids postcode, but adding it – an obvious next step – would introduce a proxy for ethnicity and income and risk indirect discrimination under the Equality Act 2010. *Socially*, pricing models trained on historical sales can entrench inequality by under-valuing property in deprived areas, so past disadvantage is reproduced as prediction (O'Neil, 2016). *Ethically*, these models learn from previous human valuations and inherit any bias in them. *Professionally*, the near-zero Rating/price correlation must be reported even though it is unflattering, and quoting the B1 prediction of £181,951 without its £24,000 error margin would misrepresent its precision, contrary to RICS transparency obligations (RICS, 2021).

Scenario 2 – expression monitoring. *Legally*, facial images processed to infer mental state are biometric data engaging Article 9 of the UK GDPR, requiring explicit consent and a Data Protection Impact Assessment (ICO, 2011). *Socially*, continuous monitoring erodes psychological safety, and expression norms vary across cultures and with neurodivergence, so a uniform model misreads some groups more than others. *Ethically*, employee consent may not be freely given, since refusal carries perceived career risk (Floridi *et al.*, 2018). *Professionally*, presenting a 72%-accurate classifier – one that misread two distressed faces as Neutral in 50 images – as reliable would breach the ACM requirement to assess and disclose harms (ACM, 2018).

4.2 D2 – Countermeasures

For Scenario 1: complete a DPIA before deployment; exclude postcode and other protected-characteristic proxies, as done here; publish prediction intervals alongside every point estimate; apply explainable-AI methods such as SHAP so a valuation can be justified feature by feature; and audit periodically for disparate impact across areas.

For Scenario 2: obtain explicit, freely given and withdrawable consent with a penalty-free opt-out; report only aggregates, as the C2 design does; retain a human in the loop so a flag triggers a conversation rather than an action; publish the measured accuracy and weakest class to the workforce; delete raw images once the expression score is derived; and review error rates across demographic groups rather than relying on one headline figure.

4.3 D3 – Intelligent Systems for the Common Good

In Scenario 1, transparent pricing models can democratise market information, giving first-time buyers analytical capability previously confined to institutional investors, while aggregated signals help local authorities identify gentrification and target affordable housing. The Rating finding shows the wider value: data science can reveal that an established professional practice does not track the outcome it claims to.

In Scenario 2, the same technology deployed with consent supports early detection of mental health deterioration in clinical settings, and in education can identify distressed students for pastoral support (D'Mello *et al.*, 2017). What separates beneficial from harmful use is not the algorithm but who holds the output, whether participation is genuinely voluntary, and whether the error rate is disclosed.

5 Section E: Conclusion and Personal Reflections

5.1 E1 – Conclusion

In Scenario 1, cleaning proved to be where most of the value lay: nine `Parking` values stored as words would have been silently destroyed by a naive numeric conversion, and since `Parking` is the strongest price predictor ($r = 0.691$), that defect alone would have degraded every downstream model. Multiple linear regression explained 76–81% of price variance with a mean absolute error near £24,000 – sufficient for indicative guidance, not for setting an asking price unaided.

Classification of Rating exposed the limits of the data. With grade 5 holding 56% of records and three grades holding fewer than ten examples, neither kNN nor SVM beat the majority-class baseline on the raw eight-grade target. Re-posing the task as three bands raised macro-F1 from 0.323 to 0.503, showing that matching the question to the resolution the data supports can matter more than the choice of algorithm. Clustering found two balanced market segments; t-SNE separated them more attractively, but PCA was selected because only its axes can be explained to a client.

In Scenario 2, both detection methods clearly beat chance: reference matching against a single exemplar reached 66.0% accuracy and the direct classifier 72.0%. Neutral was the weakest class for both, and they failed in opposite directions – reference matching never issued a false Neutral but missed seven of nine true ones, while the classifier misread two genuinely distressed faces as Neutral. That error shaped the C2 design, where aggregate reporting, dual level-and-trend rules and mandatory human review follow from the measured limitations rather than being generic caveats.

The overriding lesson is that the headline metric is rarely the honest one. Accuracy flattered models that had learned only to repeat the majority class; R^2 was highest on the split with the least reliable test set; and a bar chart of predicted expressions would have looked plausible while concealing seven missed Neutral faces.

5.2 E2 – Personal Reflections

The most instructive problems here were not the modelling steps but the points where code ran successfully and produced wrong answers.

The clearest was the `Parking` column. My first cleaning routine converted it to numeric without error and reported no missing values afterwards. Only when I printed what the conversion had *changed* did I find that nine values reading `One`, `Two` and `Three` had become NaN and then been overwritten by the median. Nothing failed; the data was simply wrong. Making cleaning routines report what they alter is now a habit, and it caught a second defect – the column detector searched for "year" and never matched `Build`, so year of construction was silently dropped from the feature set.

Evaluation taught a second lesson. My initial kNN chose k by test-set accuracy, which I had not recognised as leakage until cross-validation disagreed with it ($k = 11$ against $k = 7$). Likewise I first judged the classifiers on accuracy and concluded kNN was better, until macro-F1 showed it scoring 0.00 on five of the eight grades. Choosing a metric is a modelling decision, not a reporting formality.

A third came from my own C2 simulation. I wrote the expression-probability weights as a positional list, but the labels were sorted alphabetically, so the "calm" weight landed on Angry and the stress model ran backwards – employees scripted to deteriorate appeared to improve, with entirely plausible-looking output. Keying the weights by name removed the whole class of bug, teaching me to prefer constructs in which the mistake cannot be expressed.

Technically I gained fluency in pandas, in building and evaluating supervised and unsupervised models with scikit-learn, and in applying pre-trained transformers through Hugging Face – including extracting intermediate embeddings for one-shot matching rather than only consuming the final classification head. Dependency management proved as consequential as modelling: an early attempt to use DeepFace failed because it required TensorFlow, which had no build for my Python version, and I resolved it with a PyTorch-only Vision Transformer.

Most valuable was learning scepticism about my own results. The habit I am taking forward is checking every plausible answer against a baseline, because a model producing sensible-looking numbers while learning nothing is far more dangerous than one that visibly fails.

References

ACM (2018) *ACM Code of Ethics and Professional Conduct*. Available at: `https://www.acm.org/code-of-ethics` (Accessed: 10 August 2026).

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